

CHAPTER 6

Oh, shoot! I forgot about that! Estimating the influence of uncontrolled variables

Fools ignore complexity. Pragmatists suffer it.

Some can avoid it. Geniuses remove it.

Alan Perlis

Introduction

Description, prediction, and control are often touted as the hallmark levels of scientific endeavors (e.g., Cooper et al., 2020; Moore, 2010). When researchers first begin to study some phenomenon, they often start by describing what exactly it is they observe and the relations between those observations. Once they have gathered enough observations such that the relations between observations become reliable, they can begin to predict when, and under what conditions, certain events will occur. But, as reminded via the often-noted phrase “correlation does not equal causation,” it is possible to predict that something will occur with tolerable accuracy or precision without knowing whether one event truly causes—plays a controlling role—in the presence or degree to which another event occurs.

To this point in the book, we have primarily focused on various ways of describing our observations via the set of verbal behaviors called statistics. In these realms, we learned about the unique and precise impact that numerical descriptions of our observations can have on listeners. We also discussed measures (descriptions) of central tendency to efficiently and precisely communicate general patterns of behavior-environment relations as well as measures (descriptions) of variability within our datasets to efficiently and precisely communicate how variable behavior-environment relations might be for a client or participant. Finally, in the last chapter, we discussed how behavior analysts can describe the effect an intervention has on behavior and

how behavior analysts can use that information to predict behavior change following the same intervention for one of their clients or participants. In this chapter, we turn our attention to some of the complexity of human behavior and that third level of science—control.

Situating this chapter in the broader analytic landscape

What does it mean to control something?

The function of the verbal stimulus “control” likely differs for different people based on their learned history. But, at least within the behavior science community, “control” has been used in at least three ways (Table 6.1). The first is within a larger philosophy of science context (top row, Table 6.1). To theoretically account for the control of behavior means that we can use past behavior analytic research to offer a plausible explanation for when we might observe an increase, decrease, or the maintenance of some pattern of responding. Here, we are talking about hypothetically controlling two broad things: (1) the

Table 6.1 Three uses of “control” in behavior science.			
Area	Description	Degree of precision	Degree of certainty
Theoretical	Generalized description of behavioral processes; a position is taken on the total set of variables that can play a causal role in a behavior occurring.	Low. Typically vague and broad, meant to describe the generalized processes of environment-behavior relations to plausibly account for observed patterns of behavior.	Low-to-moderate. Typically describe how an operant/ respondent approach <i>might</i> account for behavior. Might be the most plausible explanation among available alternatives but certainty is unknown until tested directly.
Practical/ experimental	Explicit description of environmental variables assumed to play a causal role in behavior occurring, and exactly how manipulating those variables will lead to behavior change.	Moderate-to-high. Typically involves visual display of how an environmental manipulation leads to an increase, decrease, or maintenance of responding. Quantitative descriptions make the behavior-environment relations direct and explicit.	Moderate-to-high. Certainty increases with more replications of similar directional and degree of differences in behavior between the experimental and “control” conditions.
Analytic/ statistical	Explicit description of potential confounding variables and predictions around the influence those variables have on behavior. Used with variables that cannot be manipulated empirically.	Moderate-to-high. Typically involves mathematically precise claims about the size and direction of influence that an uncontrolled variable might have on responding.	Low-to-high. Certainty increases with (1) more data; (2) more accurate data; (3) greater coverage of the range the uncontrolled variables can take; and (4) uncontrolled variables have logically known or plausible functional relation with behavior.

presence of a response class and (2) the environmental conditions that theoretically are likely to influence that response class. This often makes theoretical descriptions vague and broad (as they are meant to be). And, though they might be the most plausible explanation among available alternatives, certainty around their accuracy is unknown until tested directly.

Theoretical extensions become practical when we get to the second use of the term control in behavior science: demonstrating the experimental control of a specific response topography (second row, [Table 6.1](#)). Note that we have now expanded the precision (and subsequent set) of things that a behavior analyst is controlling. Practically, the behavior analyst must have identified and controlled the environmental conditions suspected to influence responding (e.g., implement a behavioral intervention with high fidelity), they must have observed a change in responding as predicted by those environmental conditions (e.g., reduction in challenging behavior), and they must explicitly create a “control condition” to demonstrate that the pattern of behavior change does not occur in the absence of the independent variables (IVs) being tested (e.g., baseline condition; “no-treatment” control condition).

Experimental control is obviously the ideal level of control for making claims about causal relations between the environment and behavior. But, unless you have some magic sauce the rest of us do not, as mere humans we are unable to control everything in the universe when conducting an experiment. And, in applied settings, we have more limited control of variables that may influence behavior relative to laboratory settings. Nevertheless, many questions arise where we might have or are able to collect some data, we want/need an answer to inform a decision, and the experimental control of relevant variables is not practically or physically possible. For example, questions that fit this description might be: Is a behavior analyst better at developing programs for clients of a certain age and with a certain behavioral profile? How does the number of programs we are working on with Lingyun influence his rate of acquisition? Are we truly seeing more progress from clients at Clinic A compared to Clinic B, or do they have truly different types of clients? And, how do different components of our current staff training translate to on-the-job performance one week, one month, and one year after training? And, how does that interact with who their supervisors are?

In the abovementioned situations, we, fortunately, don't have to throw our hands up in ignorance, flip a coin, and let random chance determine how we proceed. Less often discussed (or used) in published articles is how behavior analysts might still get an answer to their questions when they have some data on a topic but are unable to control for all potential variables that might help answer our question. For example, variables with a known influence on human behavior and that cannot be manipulated or controlled experimentally include age, preexperimental or preintervention learning history, verbal repertoire, sociocultural background, and physiological states and conditions (e.g., comorbid disorders, currently sick). In these situations, analytical or statistical control is often the most effective (and only available) tool in behavior analysts' tool belt. This chapter focuses on this last area.

How might we control for nonempirically controlled variables?

At least two approaches can be used to handle nonempirically controlled (NEC) variables. One method is to simply ignore them. To the extent that none of those variables play a significant enough role in an experiment or intervention, ignoring NEC variables is unlikely to impact the patterns of environment-behavior relations you observe and the conclusions you can draw. However, ignoring potential confounding variables has the drawback that—without measuring or manipulating them directly—you can never be sure as to whether they did influence behavior outside your awareness. Further, if one of those NEC variables changes substantially at a future point in time, you likely cannot make predictions about how behavior will change. This may create significant limitations to the results of your experiment or lead to shifts in behavior that teachers, therapists, and parents were not expecting.

A second method for handling confounding variables is to simply collect ongoing and accurate measurements of the naturally varying presence of potentially influential variables across an experiment or intervention. Over time, as the amount of data collected increases, the behavior analyst can use mathematical techniques to identify the likely size, direction, and pattern of the relationship between the potentially influential variables and the behavior of interest (bottom row, [Table 6.1](#)). It is this last method of controlling for potentially influential variables that is the focus of this chapter.

Summarizing our situation

Biological life involves an intricate and complex set of interactions between the totality of organisms embedded within a dynamically changing environment. Though behavior analysts do their best to control all factors that can plausibly influence a response class, experimental control is not always possible for all variables' known (or with probable) relation to behavior. But just because we cannot control for these variables experimentally does not mean we cannot measure, track, and analyze the extent to which these variables are likely to influence behavior. In this chapter, we discuss how behavior analysts can control (analytically) for these (experimentally) uncontrolled variables—even if it is less ideal than experimental control. To do this, we get to start talking a bit more seriously about models of behavior-environment relations. Such talk shouldn't be cause for worry because, similar to the term "statistics," you are likely already playing with models of behavior-environment relations even if you have not talked or thought about it in that way.

Models of behavior

At the most basic level, models can be defined as a representation of a thing, physical structure, or physical process ([Merriam-Webster, 2022](#)) that are "smaller" than the thing they are meant to describe. Here, "smaller" might be along the lines of physical proportions (e.g., a model plane that won't quite get us to the Bahamas) or the inclusion of fewer details (e.g., the three-term contingency). Models can take on many different forms and the behavior of modeling may have many different functions (for a full review of this topographical and functional landscape for models in behavior science). In behavior analysis, models are often used to efficiently communicate about known or likely behavior-environment relations without having to include data and descriptions about every detail of the environment antecedent and consequent to the behavior of interest for all observations we have ever made.

Practicing behavior analysts already use models of behavior. For example, consider the statement, "If you haven't paid attention to Bamidele in a while, she is likely to start acting silly until you give her attention." This statement is a model because the speaker is describing a physical process (positive reinforcement) and the speaker is using

fewer details (21 words in total) than physically occurred on the numerous occasions where observations were made and the data were collected to derive that functional relation. More generally, the previous statement is one example of the model tacted as a three-term contingency (i.e., $S^D \rightarrow B \rightarrow S^R$) where a physical system is being described in fewer details than the totality of physical events it describes. Often, these models used and contacted by behavior analysts are textual-vocal or, perhaps, graphical models (Moxley, 1982).

You may recall from Chapter 1 that statistics are a branch of mathematics whose topic is the collection, analysis, interpretation, and presentation of aggregate quantitative data. Models simply create a framework for analyzing, interpreting, and presenting aggregate data. Statistical models are, therefore, similar to textual-vocal models in that they both aggregate patterns of behavior-environment relations and allow the model user to understand how they might predict and control behavior by changing the environment. The difference is that one of those models uses words to aggregate those patterns (textual-vocal) and the other uses mathematics and equations (statistical).

Statistical models have several practical advantages over textual-vocal models (Dallery & Soto, 2013). These include greater precision in describing operant and respondent contingencies; more precise predictions about the relations between behavior and the environment; the ability to falsify claims about behavior-environment relations; better allow researchers and practitioners to unify observations of diverse phenomena; and can easily include and analyze the influence of dozens or hundreds of variables on behavior. Stated differently, quantitative models of behavior-environment relations make more precise predictions about what might happen based on our assumptions of everything we believe is controlling behavior (and have data for). In turn, we can examine how close our predictions are, identify where we missed the mark, and improve our predictions and understanding about the variables controlling behavior. Similar to many topics in the book thus far, these models can come in many flavors.

Regression models

Beginning with the familiar

The generalized matching equation is one example of a popular statistical model of behavior-environment relations in behavior analysis

(Baum, 1974; McDowell, 1989). The prototypical matching law experiment involves exposing a rat or pigeon to dozens of experimental sessions under varying contingency arrangements (Fig. 6.1). In each session, the organism is placed in an operant chamber where two concurrent variable-interval (VI) schedules are in effect. For example, one lever or key might be a VI-60-second schedule, whereas the second lever or key might be a VI-300-second schedule. In this example, the

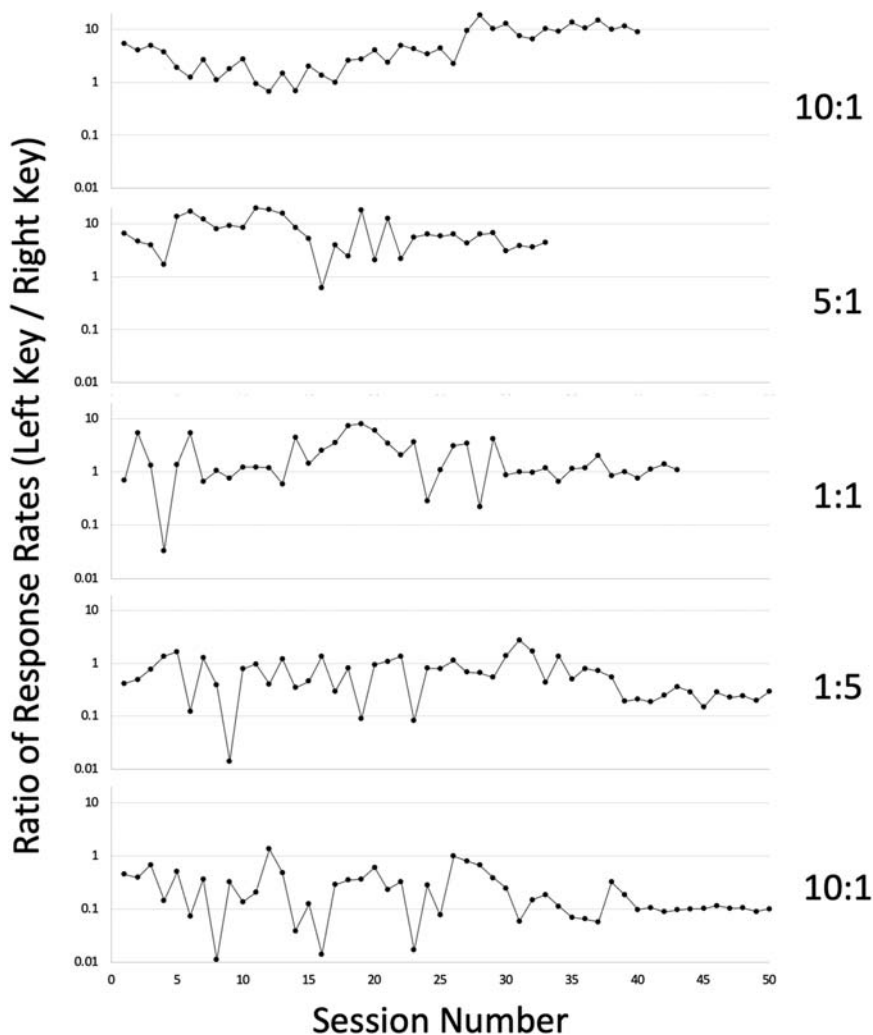


Figure 6.1 Example datasets from a prototypical matching law experiment. Each panel shows the relative response rates from one concurrent schedule ratio. Note: each data point represents an aggregate of responding for that individual for that single session (i.e., it is a descriptive statistic).

first lever or key allows the organism to contact five times the amount of reinforcement compared to the second lever or key¹. Stated succinctly, the left:right lever ratio of reinforcement is 5:1. The rat or pigeon then contacts many sessions with this concurrent schedule arrangement until responding is stable. Once stable, the ratio of left:right programmed reinforcement would change to, say, 10:1. The organism contacts this arrangement until responding is stable, the ratio is changed again to, say, 1:1 until responding is stable, then 1:5 until stable, and finally 1:10. At the end of all this, we can calculate the correct measure of central tendency (Chapter 3) to get a single data point that represents the rate of stable responding in that condition (marker at far right of each panel in Fig. 6.1). Note here that we are already at one level of aggregation per condition—statistics!

Fig. 6.2 shows how the aggregated data from such an experiment is often presented for analysis and interpretation. Here, the behavior analyst takes those calculated average rates of responding and then uses computer software (e.g., Excel, Prism, Python, R) to create a scatterplot of the data. Once created, the behavior analyst can add the “line of best fit” to those data². If you recall from the math classes you may have taken when you were younger, the equation for a line is:

$$y = mx + b. \quad (6.1)$$

Here, y are the values corresponding to the y -axis, x are the values corresponding to the x -axis, m is the slope (i.e., how much y changes for every one unit change in x), and b is the y -intercept (i.e., the y -value when $x = 0$).

Getting the line of best fit is useful because the generalized matching equation is a linear model (i.e., uses the equation for a line—Eq. 6.1). The only difference is that we swap out the dependent variable (DV) y for the ratio of responding (B_{left}/B_{right}) and we swap out the IV x for the ratio of reinforcement (SR_{left}/SR_{right}). The resulting equation

¹ With a VI-60-second schedule the organism will likely contact 60 reinforcers per hour (1 reinforcer per minute). With the VI-300-second schedule (1 reinforcer every 5 minutes), the organism will likely contact 12 reinforcers per hour ($60/5 = 12$). Left key:right key = $60:12 = 5:1$.

² For those unfamiliar with how to do this in your favorite data software program, a quick Google search of “How to add a linear regression line in [program]?” should get you what you need (while swapping out [program] for the program you use; e.g., Excel, Prism). For the generalized matching equation, specifically, see the excellent tutorial by Reed (2009).

A	B	C	D
SR+ Left / SR+ Right	Resp. Left / Resp. Right		
10	9.7	0.97	Slope
5	4.9	0.04	y-Intercept
1	1	0.9985	r^2
0.2	0.24		
0.1	0.11		

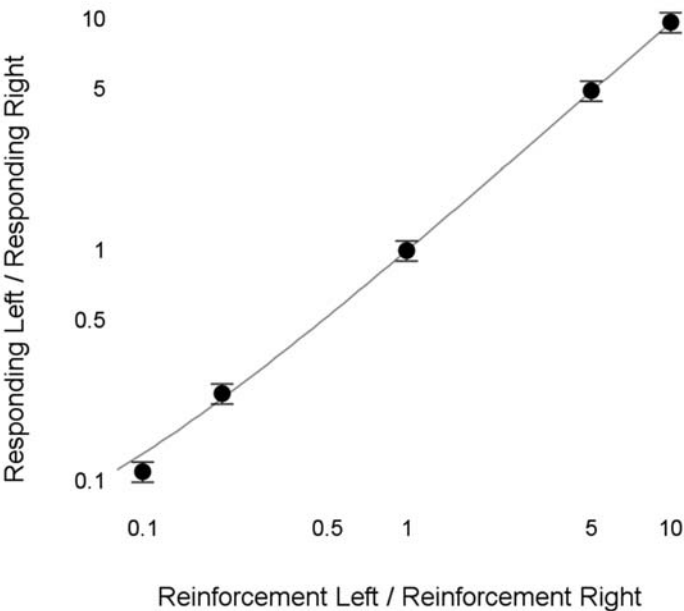


Figure 6.2 The spreadsheet shows the average rates of stable responding aggregated across the five conditions in Fig. 6.1. The graph shows one common method to visually display those data: a scatterplot with the line of best fit determined via linear regression.

then becomes:

$$\frac{B_{left}}{B_{right}} = m \left(\frac{SR^{+}_{left}}{SR^{+}_{right}} \right) + b. \tag{6.2}$$

This makes m and b interpretable from a behavioral perspective. m (the slope) describes how sensitive the organism is to detecting changes in reinforcement ratios (as measured by response ratios)³. And, b describes a

³ Readers interested in what all goes into organismic sensitivity to changing reinforcement schedules might get started with Davison and Tustin (1978), Madden and Perone (2013), and McCarthy and Davison (1980). Then, let the wonderful citation train be your tour guide.

Table 6.2 Types of models and their characteristics.				
	Model type	Number independent variables	Dependent variable data type	Example from behavior analysis
1	Univariable linear regression	1	Continuous	Generalized matching equation
2	Univariable nonlinear regression	1	Continuous	Discounting, demand
3	Multivariable linear regression	2 +	Continuous	Concatenated matching law
4	Multivariable nonlinear regression	2 +	Continuous	Additive and multiplicative discounting; MPR
5	Univariable ordinal regression	1	Ordinal	Open vs. closed economy to predict PSPA Ranks; AUC _{ORD}
6	Multivariable ordinal regression	2 +	Ordinal	Economy Type and Session Setting on PSPA ranks
7	Univariable classification	1	Ordinal/nominal	Effect of newsletter on political action (Schroeder et al., 2004)
8	Multivariable classification	2 +	Ordinal/nominal	Predictors of “yes” or “no” intervention effect using visual analysis (Kahng et al., 2010)
MPR, <i>Mathematical principles of reinforcement</i> (e.g., Killeen, 1994; Killeen & Sitomer, 2003); PSPA, <i>paired stimulus preference assessment</i> ; AUC _{ORD} , <i>area under the curve using delays or probabilities ordinally</i> (e.g., Borges et al., 2016).				

general preference for one of the responses not captured by the measured reinforcement schedules⁴. The spreadsheet in Fig. 6.2 shows the slope and y-intercept for the data in the figure. Inserting those values, the researcher could say, “For every one unit change in reinforcement ratio, we expect a 0.97 change in response ratios.” More generally, we have used a mathematical model to account for (1) how changing schedules of concurrent reinforcement influence changes in behavior; (2) how sensitive this specific organism is to changing schedules of reinforcement; and (3) general preference for one response that we did not control for in our experiment.

Generalizing to a bigger picture

If you feel comfortable with what behavior analysts are doing when playing with the matching law (i.e., building a linear regression model interpretable relative to behavioral processes), you are in great shape to understand statistical models more generally. Recall that “statistics” just means the analysis, interpretation, and presentation of aggregate quantitative data. The top row in Table 6.2 shows the general features

⁴ One final technical note for the generalized matching equation is that—to get the data to be well described by a straight line—researchers take the logarithm of the reinforcement ratios and the response ratios before fitting the regression line (see Baum, 1974 for why this is useful). For our example, we simply logged the axes for similar effect.

of the type of statistical model that we call the generalized matching equation. In the generalized matching equation, we have one IV (i.e., reinforcer ratios where each plotted ratio is an aggregate from many sessions), we are predicting a continuous DV (i.e., response ratios where each plotted ratio is an aggregate from many sessions), and we use a line to describe a relationship between reinforcer ratios and response ratios. In the jargon of statistics, the generalized matching equation is a **univariable linear regression model**. “Univariable” because we have one IV, “linear” because we are using a line to describe the relationship between the IV and DV, and “regression” because the DV is continuous.

As described above, linear models are useful because they are easy to fit to our data and are also very easy to interpret. But life is complex and data often does not fall on simple lines. When the relationship between variables does not fall along a straight line, we call it (quite logically) a nonlinear relationship. And, when we are still using a single IV to describe or to make predictions about a single DV, we again use the univariable label. Thus, when we are interested in talking about a single IV describing a single continuous DV but via a nonlinear relationship, we call such statistical models **univariable nonlinear regression models**.

The second row in [Table 6.2](#) highlights two nonlinear statistical models in behavior analysis: discounting (e.g., [Rachlin, 2006](#)) and demand (e.g., [Hursh & Silberberg, 2008](#)). In both models, there is one IV (delay, probability, effort, or social relatedness for discounting; price to contact a reinforcer for demand) and a continuous (or near continuous) DV (reinforcer value for discounting; reinforcers consumed for demand). Also for both models, the relationship between the two variables is nonlinear ([Fig. 6.3](#)). For discounting, the relationship between the IV and DV is often hyperbola-like (e.g., [Rachlin, 2006](#); left panel [Fig. 6.3](#)). For demand, the relationship between the IV and DV is often exponential (e.g., [Hursh & Silberberg, 2008](#); right panel [Fig. 6.3](#))^{5,6}. Bringing it all

⁵ For the adventurous, one equation that describes the nonlinear relationship between IV and DV in discounting is $V = A/(1 + \gamma X)$ where V represents reinforcer value (i.e., how much behavior the reinforcer will maintain), X is the value the IV takes (e.g., delay in days, odds against), and γ is a free parameter that describes how quickly the DV changes as the IV changes (i.e., how reinforcer value changes with delay, odds against, effort, or social relatedness).

⁶ For the adventurous, one equation that describes the nonlinear relationship between IV and DV in demand is $\log_{10} Q = \log_{10} Q_0 + k(e^{-aQ_0C} - 1)$ where Q represents reinforcers consumed, Q_0 represents the reinforcers consumed when the price is 0 (i.e., free), k is a constant that represents the range of consumption, and C is reinforcer cost (i.e., the IV), and a describes how quickly the DV changes as the IV changes (i.e., how reinforcer consumption changes with cost).

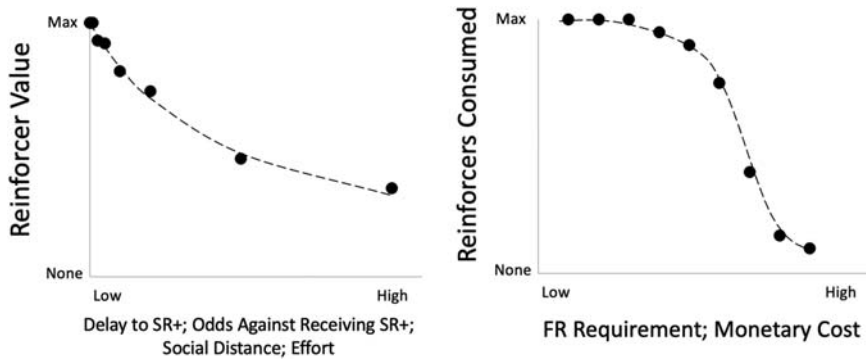


Figure 6.3 Two popular statistical models in behavior analysis that show nonlinear relationships between the independent variable and the dependent variable.

together in the succinct language of statistical jargon, discounting and demand are univariable nonlinear regression models.

The generalized matching equation, discounting equation, and demand equation are all examples of univariable statistical regression models. They have one IV that we are using to predict or describe changes in one DV. But most behavior is likely influenced and controlled by many different variables, that is, the multiple control of behavior is likely the rule rather than the exception (Michael et al., 2011; Skinner, 1957). Statistical models of behavior-environment relations that describe how multiple variables' influence behavior are called (quite logically) **multivariable models**. And, as before, we can use either a simple line to describe that relationship (linear model) or use something other than a simple line to describe that relationship (nonlinear model).

The third row in Table 6.2 highlights the features and one example of a **multivariable linear regression model** in behavior analysis: the concatenated matching law (e.g., Davison, 1982; Davison & McCarthy, 1988; Hunter & Davison, 1982). Recall that the generalized matching equation predicts that the ratio of rates of two behaviors is equal to the ratio of reinforcement rates contacted by each behavior, with a free parameter slope accounting for sensitivity to changes in reinforcement rates, and the free parameter y -intercept describing bias for one response or the other. But, we also know that behavior is influenced by the *amount* of reinforcer at each reinforcer delivery as well as the *immediacy* with which the organism contacts reinforcement after responding (e.g., Rodriguez & Logue, 1986; Vollmer & Bourret, 2000).

The concatenated matching law attempts to account for this by adding amount (A) and immediacy (I) into the statistical model:

$$\log\left(\frac{B_{left}}{B_{right}}\right) = m \times \log\left(\frac{R_{left}}{R_{right}}\right) + a \times \log\left(\frac{A_{left}}{A_{right}}\right) + i \times \log\left(\frac{I_{left}}{I_{right}}\right) + b. \quad (6.3)$$

Just as before, we have the rate of reinforcement (R), sensitivity to changing reinforcement rates (m), and bias (b) in the equation. We have also now added the ratio of amount (A), sensitivity to changing ratios of reinforcer amounts (a), ratio of reinforcer immediacy (I), and sensitivity to changing immediacy ratios (i). Multiple variables (i.e., rate, amount, and immediacy = multivariable) aggregated over many sessions (i.e., statistical) are being used to predict via a linear equation a continuous DV (i.e., ratio of behavior). Succinctly via statistical jargon: a multivariable linear regression model.

The fourth row in [Table 6.2](#) shows the features and some examples of ***multivariable nonlinear regression models*** in behavior analysis. Two recently explored models within discounting sought to understand how the delay (one IV) and probability of (a second IV) an outcome combine to influence reinforcer value within discounting studies (i.e., multivariable; e.g., [Cox & Dallery, 2016](#); [Vanderveldt et al., 2015](#)). And, just as with discounting previously, there is a nonlinear shape to the relation between reinforcer value (i.e., a continuous variable—regression) with delay and probability. We will leave it to the curious reader to check out those papers if interested in the model details. But there are three main points here. First, multivariable nonlinear regression models are doing the same thing as all the above regression models, just using two or more IVs and an assumed nonlinear relationship between the IVs and a continuous DV. Second, behavior analysts already use these models to quantitatively describe the multiple control of choice, as constrained by known behavioral processes. Third, similar to the regression models above, the estimated parameters associated with each IV gives you a precise estimate of the degree of influence that variable has on behavior (i.e., it is behaviorally interpretable).

Rows 5 and 6 of [Table 6.2](#) help us round out the types of regression models behavior analysts are likely to use. As discussed in most chapters to this point, many different types of data exist and each data type

should be handled differently based on the logic of numbers. Modeling is no different. Above we talked about regression modeling when our DV is a continuous data type. But sometimes the measure of behavior we want to model is ordinal. For example, item rankings from preference assessments are ordinal and we would need to use a different equation to model these data (e.g., [Liu et al., 2017](#)) compared to a continuous DV (e.g., rate of responding). Nevertheless, just like continuous data, ordinal regression models can be linear or nonlinear, and ordinal regression models can involve one predictor variable (univariable) or multiple predictor variables (multivariable). One final note here is that using regression models for ordinal data only makes sense if you have enough levels of your ordinal data such that it starts to look and smell similar to continuous data (e.g., the DV can take dozens or hundreds of values). If your ordinal DV does not have enough levels to dress like a continuous variable, you can pivot to models designed for discrete nominal data: classification models.

Classification models

As was highlighted in previous chapters, much of the data that behavior analysts collect are not continuous in nature or we might not be able to transform it to a continuous data type. In these instances, we build what are referred to as ***classification models***. This makes logical sense if you step back and think about it. If we want to predict, say, your favorite color, we would be predicting a category or label or class of the total set of stimuli that humans tact as “colors.” Examples from applied behavior analysis might include predicting whether a client needs comprehensive or focused intervention based on the totality of their presenting symptoms, classifying the function of a behavior based on all the environment-behavior data collected, predicting whether a stimulus will or will not function as a reinforcer (or punisher), classifying whether or not staff implemented a step of an intervention procedure correctly, predicting whether or not staff are likely to resign, or classifying whether or not a change in behavior occurred based on visual analysis.

Approaches to classification modeling can be discussed relative to at least two characteristics of the things we have collected data on. First, how many IVs do we have that we are using to describe or predict our DV. As with regression models, the same jargon is used to describe whether we are using one IV (***univariable***) or two or more IVs

(*multivariable*) to describe or predict their influence on a discrete DV. Second, whether we are attempting to predict if responding falls into one of two different categories or if responding falls into two or more categories. When our DV has only two (*bi*) categories or levels, it is called **binary classification**. When our DV has three or more categories or levels, it is referred to as **multinomial classification** (“multi” meaning many and nominal meaning “names”; multinomial = many names).

The seventh row of [Table 6.2](#) highlights the characteristics of univariable classification models. Let’s start simply with binary classification. The rightmost column of row 7 shows one example from the *Journal of Applied Behavior Analysis* where researchers were interested in examining the relative influence of one IV on a binary ordinal DV. [Schroeder et al. \(2004\)](#) evaluated whether sending business owners a newsletter twice a week (one IV) led the business owners to take political action on environmental issues (“yes, they took action” or “no, they did not”—binary classification). This is also an ordinal DV because “yes, they took action” represents more behavior than “no, they did not take action,” but we do not know how much more behavior they took. So we have a univariable binary classification task in front of us.

Arguably, the most popular binary classification model is called a **logistic regression model**. We know, we know. After all the hoopla above around defining regression vs. classification, it is likely very confusing that the first classification model we talk about is labeled a “regression.” But it makes some sense when we dig into it and we see the output of the model. And it also sets us up for what the many classification models are doing that you will likely use or encounter.

[Fig. 6.4](#) shows some data and a visual for what we are doing with the logistic regression model. Our measure of behavior can be either a “0” (did not occur) or a “1” (did occur). Sticking to the univariable model, we also have one IV (e.g., newsletters per week, reinforcers delivered per hour). Because the DV can only take on a “0” or “1” value, a linear function would not make much sense as it could predict values less than “0” or greater than “1” (dashed line in [Fig. 6.4](#)). The potential predicted values in the middle are not as concerning because of the magic of numbers and of which logistic regression solves for us quite handily.

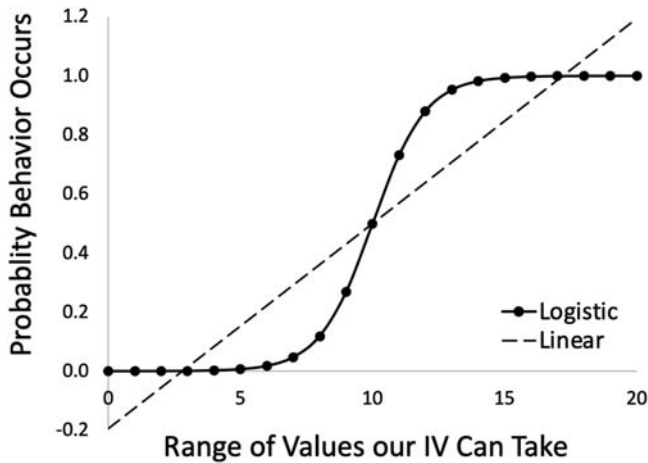


Figure 6.4 Visual of the logistic regression output.

The solid line with markers in Fig. 6.4 shows the output of a logistic regression model⁷. Because of the math behind logistic regression, the output values range only between “0” and “1” (solid line in Fig. 6.4). As can be seen in the plot, as the x -values decrease toward zero, the predicted y -values approach “0.” And, as the x -values increase toward the maximum value our IV can take (i.e., 20 in this example), the predicted y -values approach “1.” In turn, we can interpret the output as the probability that the observation will be a “1” based on the value of whatever is represented by our x -axis (i.e., the value of our IV)⁸. That’s the magic, because we have constrained our output between 0.0 and 1.0, we can interpret the numerical output as a probability. So, as the output gets closer to 1.0, we can interpret it as a higher probability that we will observe whatever ordinal or nominal label we gave to “1.” Translating to behavior analytic contexts, a “1” might mean “the individual emits the target behavior.” So, increasing the amount of the IV increases the probability we observe the target behavior. Quite logical! And, because the output of our model can be interpreted as a

⁷ For the reader interested in greater mathematical technicalities, we have shifted our x -axis range for ease of demonstration in Fig. 6.4 as IV values for most behavior analytic interventions and experiments cannot be negative. Technically, the 0.5 output is centered at $x = 0$ for the generalized logistic regression case. And, software packages handle the transformation behind-the-scenes to make the output interpretable for you.

⁸ If it makes sense based on the behavior-environment relations you have measured, you can reverse this function so that probability approaches 1.0 as x approaches zero and probability approaches 0.0 as x approaches the max value it can take.

probability, it can take on any real value between 0.0 and 1.0—a regression! Also, quite logical!

So, logistic regression models (and most classification models) give us a probability that a “1” is observed. The next step is to turn that continuous probability value into a discrete prediction of a “1” or a “0.” This is done quite simply by choosing a cutoff probability. Then, any predicted probability above that cutoff value would be classified as a “1” and any predicted probability below that cutoff would be classified as a “0.” The default cutoff is often 0.5 for most statistical software programs and programming packages. If you’re thinking, “That seems quite arbitrary,” then you are 100% right! There is no inherent reason the cutoff point has to be 0.5. You could set it to 0.05, or 0.6, or 0.923487364587643756324. Pick your poison! In the following section, we talk a bit more about how you might change that cutoff value. The point here is that logistic regression uses our IV(s) to make a prediction between 0.0 and 1.0 that your observation falls in one of two categories. And, you can change the cutoff point that leads to discrete labels based on your data and what you’re trying to model.

What happens if we have more than one category value that our DV can take? Well, if you followed what we are after with a binary classification task, then you are in great shape as we just do the exact same thing with a twist. And, that twist often comes in one of two flavors: cherry or purple-blast. Only kidding, they are one-vs.-rest or one-vs.-one. [Fig. 6.5](#) shows what we are after with each of these approaches. In the one-vs.-rest approach, we create one binary classification model for each label that shows up in our dataset. So, if there are three values our DV takes in our dataset (e.g., classifying errors as indicative of molar win-stay, molecular win-stay, or position bias; see [Grow et al., 2011](#)), we create one classification model of whether the DV is molar win-stay or not, one classification model of whether the DV is molecular win-stay or not, and one classification model of whether the DV is position bias or not. The output of each of these models is the probability that the observation is that categorical value. And, to make a final choice, we can simply look at which of the three models has the highest probability, and that becomes our guess.

One-vs.-one multinomial classification also leads us to create multiple models and pick the one with the highest probability. The difference here is that, rather than building one model for each discrete

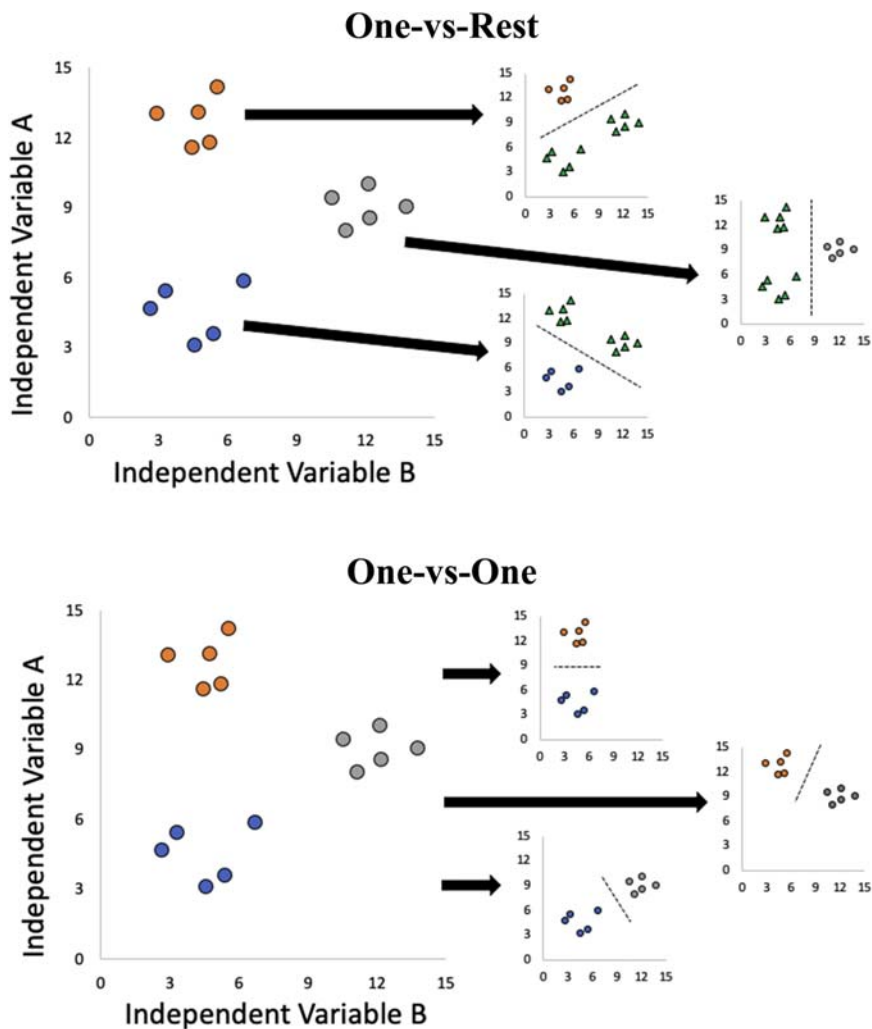


Figure 6.5 High-level overview of what multinomial classification models are doing. Please check the online version to view the color image of the figure.

value our DV takes, we create a model for each pairwise combination of values that our DV takes (lower panel in Fig. 6.5). Using the molar win-stay, molecular win-stay, or position bias example from above, we would create one model each for molar win-stay-vs.-molecular win-stay, molar win-stay vs. position bias, and molecular win-stay vs. position bias. We can then count the number of pairwise combinations that each category “wins” and, if there’s a tie, the category with the

higher probability in the head-to-head model is predicted as the “correct” category label. Easy as pie!

A question that logically follows the preceding two paragraphs is when we might want to choose the one-vs.-rest or the one-vs.-one approach to multinomial classification. The answer here really comes down to what your data looks and feels like. One-vs.-rest is really nice to use when the measures of each DV value are fairly distinct when we plot them relative to interactions between our IVs (e.g., how they are plotted in [Fig. 6.5](#)); we have pretty different amounts of each of the DV values in our dataset (aka we have an *imbalanced* datasets); or there are a lot of values that our DV takes in our dataset. To drive that last point home, say there are 10 values that our DV takes. One-vs.-rest would require our computer to create 10 models, whereas one-vs.-one would require 45 models (remember from Chapter 2, $C(n, r) = \frac{n!}{r!(n-r)!}$). This can become computationally time consuming if the number of possible values gets large enough. One-vs.-one is great when there is quite a bit of overlap in our DVs when plotted relative to IVs; we have a similar amount of each of the DV values in our dataset (aka a *balanced* dataset); and there are few values the DV takes.

To close out this section, it is important to note that we briefly discussed logistic regression as one approach to modeling discrete DVs that are either binary or multinomial. As with much of what we have covered throughout the book, this is only one of the many possible approaches to building classification models that exist in the wild. To help build some intuition around what’s possible, consider again [Fig. 6.5](#). In each model, we are essentially trying to figure out the best way to draw a boundary that separates the two or more categories we are interested in differentiating. As you can likely imagine, there are thousands, maybe millions of different ways we can draw the dashed separation boundary. And we don’t always have to use a line; we could draw a nonlinear “squiggly” boundary, or we could draw a circle around one dataset, or an oval, or a triangle, and so forth. The limit to how you think about your data really comes down to your own creativity and ability to implement your idea in whatever software program you have access to.

Hopefully, by this point in the chapter, you have a good sense of the basics behind how to model discrete ordinal or nominal data. If your

DV is multinomial, choose either the one-vs.-rest or the one-vs.-one strategy based on the characteristics of your data. Convert your data to binary format where the “1” and “0” make logical sense based on your DV. Choose the model you want to use (e.g., logistic regression) from the options available in the software package you prefer. Pour yourself a cup of coffee, mineral water, or your favorite beverage. And, let the software do the heavy lifting to fit the model and spit out an output.

Brief primer on interpreting models

To close out this section, we want to first say that the process of building accurate and reliable statistical models truly deserves a book-long treatment in itself⁹. So, though we believe that everyone should experiment and “try this at home,” be sure you run your final models and outputs past someone with some experience in this area before you go off publishing and making claims that are...not quite right. To give you some headway in this direction, we’ll close out this chapter by briefly reviewing what to do with the output that your statistical software spit at you while you were away pouring a cup of Joe. This comes in two phases. First, we can ask how well our model describes our data or makes predictions. Second, if our model performs well, we can ask how much each IV influences our DV.

How good is this model, exactly?

The first step to evaluating how good our model is involves analyzing loss/fit¹⁰ metric(s). A loss metric is pretty much exactly as it sounds. We have some kind of metric—a measure—of the loss of information that comes by way of using a model for description and prediction as opposed to using the raw data. Stated in a more long winded fashion, the most accurate description of our data is the raw data itself because it is the exact thing we want to describe and predict. But using only raw data comes with all the baggage and imprecision we have discussed to this point in the book. When we create a model of our data, it is unlikely to perfectly fit the raw observations we have made. It’ll be “off” in some way, shape, or form. As a result, we lose information

⁹ Psst...hey, Jason! What are you doing next year?!?!

¹⁰ For the modeling pros out there, we appreciate that some people distinguish loss and fit metrics. However, some of the arguments put forth on this distinction are not always convincing and often feel like splitting hairs, we left decided to keep things simple for the sake of an introduction to this material.

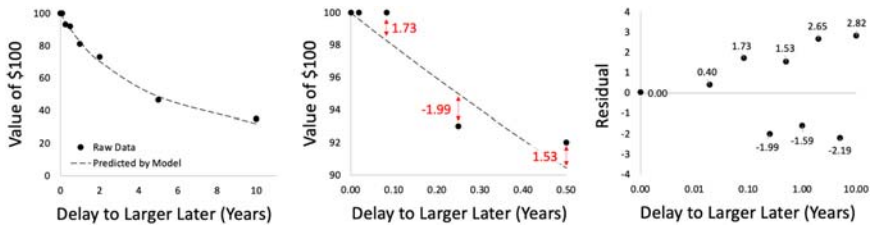


Figure 6.6 Plots showing how we derive loss metrics for regression analyses via an example of delay discounting.

that was contained in the raw data with the trade-off being increased precision in both the predictions we can make and the estimates of the influence of our IVs on behavior.

Fig. 6.6 shows a visual of what we mean here via an example of delay discounting. The leftmost panel shows a prototypical discounting dataset where we have the measured indifference points for a human, rat, or pigeon participant. The middle panel of Fig. 6.6 zooms in to look at just the first five indifference points lasting up to a 0.50 years delay to the larger later reinforcer. Here, it is easier to see how “off” our model is—the gap or the difference—between the observations we made and what the model is predicting (denoted by the arrows). Formally, these are called **residuals** and we can calculate the residuals by subtracting the model’s predicted value from the observed value. Once we calculate these residuals for each observation in our dataset, we can measure how much information we lost by using our model instead of the raw data: loss metric.

The far right panel of Fig. 6.6 shows what is called a residual plot. A residual plot graphs the calculated residuals (y-axis) as a function of our IV which, in this case, is the delay to the larger later reinforcer. Residual plots should look like that shown in Fig. 6.6 where they are approximately evenly distributed around 0.0 residuals and there are not obvious trends in the data. That is, the residuals from fitting our model do not systematically become more negative or more positive as our IV increases. If we do observe systematic changes in our residuals, then we likely need to improve the model before moving forward. Assuming our residuals are evenly distributed and nonsystematic, then we can move into more formal calculations using our residuals.

How well does my regression model fit my data?

Table 6.3 shows six commonly used loss/fit metrics for regression models. Most statistical software programming packages will calculate

Table 6.3 Common loss metrics for regression tasks.

Name	Steps to calculate	Mathematical notation	Common equation	Ideal use case
r^2 aka VAC	(1) Calculate the residual for each observation. Square each.	$= (y - y')^2$	$1 - \left(\frac{\sum (y - y')^2}{\sum (y - \bar{y})^2} \right)$	Linear regression only. Interested in how model compares to using the mean to describe your data.
	(2) Add up all the calculated values in step 1.	$= \sum(\text{Step 1})$		
	(3) Find the difference for each observation from the mean.	$= (y - \bar{y})^2$		
	(4) Add up all the calculated values in step 3.	$= \sum(\text{Step 3})$		
	(5) Divide the result from step 2 by the result from step 4	$= (\text{Step 2})/(\text{Step 4})$		
	(6) Subtract the result from step 5 from 1.	$= 1 - (\text{Step 5})$		
MAE	(1) Calculate the absolute value of the residual for each observation.	$= \text{abs}(y - y')$	$\frac{\sum \text{abs}(y - y')}{N}$	Linear or nonlinear regression. Interested in how “off” your model is with metric in units of your DV.
	(2) Add up all the calculated values in step 1.	$= \sum(\text{Step 1})$		
	(3) Divide the result from step 2 by the total number of observations.	$= (\text{Step 2})/N$		
MSE	(1) Calculate the residual for each observation. Square each.	$= (y - y')^2$	$\frac{\sum (y - y')^2}{N}$	Linear or nonlinear regression. Interested in how “off” your model is with greater weight for higher residuals.
	(2) Add up all the calculated values in step 1.	$= \sum(\text{Step 1})$		
	(3) Divide the result from step 2 by the total number of observations.	$= (\text{Step 2})/N$		
RMSE	(1) Calculate the residual for each observation. Square each.	$= (y - y')^2$	$\sqrt{\frac{\sum (y - y')^2}{N}}$	Linear or nonlinear regression. Interested in how “off” your model is with greater weight of higher residuals, with metric in units of your DV.
	(2) Add up all the calculated values in step 1.	$= \sum(\text{Step 1})$		
	(3) Divide the result from step 2 by the total number of observations.	$= (\text{Step 2})/N$		
	(4) Take the square root of the result from step 3.	$= \sqrt{(\text{Step 3})}$		

AIC	(1) Calculate the residual for each observation. Square each	$= (y - y')^2$	$N \log(\hat{\sigma}^2) + 2K$	Model comparisons. Use this equation only for least squares regression with normally distributed residuals. Else, log-likelihood equation is appropriate. With fewer than 40 total observations, AICc should be used.
	(2) Add up all the calculate values in step 1.	$= \sum(\text{Step 1})$		
	(3) Divide the result from step 2 by the total number of observations.	$\frac{(\text{Step 2})}{N}$		
	(4) Take the <i>log</i> of step 3 and multiply by number of observations.	$= N \times \log(\text{Step 3})$		
	(5) Multiply the number of free parameters in your model by 2.	$= 2K$		
	(6) Add steps 4 and 5 together.	$= (\text{Step 4}) + (\text{Step 5})$		
BIC	(1) Calculate the residual for each observation. Square each.	$= (y - y')^2$	$N \log(\hat{\sigma}^2) + K \times \log(N)$	Model comparisons. You want to penalize the number of parameters to greater extent. Use this equation only for least squares regression with normally distributed residuals. Else, log-likelihood equation is appropriate.
	(2) Add up all the calculate values in step 1.	$= \sum(\text{Step 1})$		
	(3) Divide the result from step 2 by the total number of observations.	$\frac{(\text{Step 2})}{N}$		
	(4) Take the <i>log</i> of step 3 and multiply by number of observations.	$= N \times \log(\text{Step 3})$		
	(5) Add up the number of free parameters in your model.	$= K$		
	(6) Multiply step 5 times the log of the total number of observations.	$= (\text{Step 5}) \times \log(N)$		
	(7) Add steps 4 and 6 together.	$= (\text{Step 4}) + (\text{Step 6})$		
AIC, Akaike's information criterion; BIC, Bayesian information criterion; DV, dependent variable; MAE, mean absolute error; MSE, mean squared error; RMSE, root mean squared error; VAC, variance accounted for.				

these for you. However, for the Excel die-hards, the second column outlines the steps to calculate the metric. And, for the mathematically inquisitive, the third column shows the mathematical notation that coincides with each step and how they combine in the fourth column to form the overall equation. Lastly, for everyone reading this book, the last column shows the typical condition under which each loss metric is likely the best metric for the job. Importantly, note that every loss metric begins by calculating the difference between what our model predicts and the actual data: the residuals. And, again—each assumes the residuals are evenly distributed around zero and nonsystematic.

Variance account for (r^2)

The top row of [Table 6.3](#) shows arguably the most common fit metric used to evaluate models in the history of behavior analysis: r^2 ; also known as variance accounted for (VAC). Intuitively, r^2 is a comparison of two descriptive models. One model is the model we built. How well that model describes our data is the top part of the equation. The second model is arguably the simplest description of our data that we can use: the arithmetic mean of our observations. How well the arithmetic mean value describes our data is the bottom part of the equation. To compare the two models, we simply take the ratio of how well our model describes the data compared to how well the arithmetic mean describes the data. More technically, we divide the sum of the squared residuals from our model by the sum of the squared residuals from the arithmetic mean¹¹. And we subtract that ratio from 1.0. Models with r^2 close to 1.0 mean that our model describes the data really well. Models with r^2 values close to 0.0 are just as good as using the arithmetic mean to describe our data (i.e., it is not a very good model). And, models with r^2 values less than 0.0 are...well...down-right terrible. For even more context, good models derived from highly controlled experimental settings in behavior analysis typically have r^2 values greater than 0.85 or 0.90. And, good models from more complex or applied contexts have r^2 values greater than 0.60 or 0.65.

VAC is so popular that we are going to give it an extra paragraph in this section around its ideal use case. The reason for this is because r^2 is an appropriate fit metric only for linear regression models and is

¹¹ Technically, we also square the residuals before comparing them. See Chapter 4 for a discussion on why. But the high-level idea is the same: we compare the squared model residuals to the squared residuals from the mean by taking the ratio between the two.

not an appropriate fit metric for nonlinear models (e.g., discounting, demand). The mathematical reasons for this are outside the scope of this book. But, interested readers should check out [Spiess and Neumeier \(2010\)](#) for a description of the history for why r^2 might historically have been inappropriately used across sciences and a demonstration for why other metrics are better for nonlinear regression models. But, in the end, r^2 is a relatively intuitive, widely used fit metric that is best suited for linear regression models.

Mean absolute error

When we have nonlinear regression models, one sensible metric of model performance is to measure how “off” our description or prediction is from our observations on average. That’s what mean absolute error (MAE) does. Just as with all regression loss metrics, we start by calculating the residuals of our model. The “absolute” bit comes from then taking the absolute value of the residuals (i.e., make them all have a positive sign). Lastly, we calculate the arithmetic mean of those absolute residuals. The really nice thing about MAE is that the metric remains in the unit of our DV. For example, if our DV is responses per minute, and the MAE for our model is 1.21, then that means that our model is off in predicting behavior by 1.21 responses per minute on average. Whether this is good or bad likely depends on the social significance of being off by 1.21 responses per minute. Regardless, if you want an easy-to-interpret loss metric in the units of your DV and have a nonlinear model, MAE is likely a good choice.

Mean squared error

Mean squared error (MSE) is another common metric used for nonlinear regression models. Just as with r^2 and MAE, we start by calculating the residuals for each observation. Next, we square each of the residuals. The idea here is to penalize residuals that are further away from the observed values compared to the residuals that are closer to the observed values. This happens because, as residuals get larger and larger, squaring them increases their impact on our loss metric exponentially. MSE is great when the MAE for several models is relatively similar as MSE can magnify subtle differences in the residual distributions to highlight models that have more residuals that are relatively large. The downside is that the loss metric is no longer in the same units as our DV making interpretation less straightforward than with MAE.

Root mean squared error

Root mean squared error (RMSE) combines some of the benefits of MSE and MAE. RMSE is calculated in the exact same way as MSE. The only difference is that, once we calculate the arithmetic mean value of the squared residuals, we then take the square root of that average. RMSE, therefore, has the benefits of penalizing larger residuals more than MAE does and also converts the final loss metric into the units of our DV for more straightforward interpretation. Thus RMSE is a great choice if you find yourself in a nonlinear regression modeling situation and you have reason to penalize models that make relatively large errors while maintaining the unit of your DV for interpretability.

Metrics that penalize model complexity

A limitation to MAE, MSE, and RMSE is that they provide no information about model complexity. As with many sciences, behavior analysts often prefer simple descriptions and accounts of behavior over complex descriptions and accounts of behavior (Occam's razor). Model complexity is typically quantified by the number of free parameters in a model. Failing to account for the number of free parameters in a model is a limitation because models with a greater number of free parameters typically are better at describing and predicting behavior because they are simply more flexible and not because they theoretically make more sense. Akaike's information criterion (AIC) and Bayesian information criterion (BIC) are two loss metrics that evaluate models on how well they describe/predict behavior *as well as* based on the number of free parameters the model has.

AIC and BIC are calculated in very similar ways¹². Both begin with the MSE (i.e., calculate residuals, square them, add them up, and divide by the number of observations). From there, both also then take the log of this value and multiply the result by the total number of observations. This left half of each equation is the portion that accounts for how well the model describes/predicts our data. The right half of each equation is the portion that penalizes each model based on the number of parameters (K) that it has. The primary difference is that AIC multiplies the number of parameters by two, whereas BIC multiplies the number of parameters by the log of the number of observations. If you started to do the math in your head, yes—when there are 100 observations, these

¹² Readers interested in the foundations underlying these equations should refer to [Burnham and Anderson \(2016\)](#).

two equations are equivalent. But, given we often have more than 100 total data points (see Chapter 7 for a more detailed discussion of what counts as an observation), BIC typically is considered to penalize the number of parameters more than AIC.

How well does my classification model fit my data?

All classification loss metrics center around whether our model did or did not predict the correct discrete category for each observation. To make this easy, we will use an example where our DV can take one of two values, “yes, behavior occurred” = “1” or “no, behavior did not occur” = “0.” Here, “1” might be a correct response during a match-to-sample task, a mand emitted, aggression occurred, and so on. In this situation, there are two possible values that our behavior can take (1 or 0) and two possible values that our model can predict (1 or 0). Consequently, Table 6.4 shows the four possible outcomes that can occur with these two combinations. Our model might predict behavior occurs and behavior actually occurred (true positive); predict behavior occurs and behavior actually did not occur (false positive); predict behavior does not occur but behavior did occur (false negative); or predict behavior does not occur and behavior did not occur (true negative).

As you probably can guess, we want the predictions from our model to land in the upper left and lower right “true” buckets. And, we want very few observations to fall into the lower left and upper right “false” buckets. There are many different ways that we can quantify the relative distribution of the predictions from our model across these four buckets. Table 6.5 shows commonly used loss metrics when

Table 6.4 Example confusion matrix when our dependent variable is binary (i.e., is one of two values).

		ACTUAL VALUES	
		Positive (1)	Negative (0)
PREDICTED VALUES	Positive (1)	True Positive (TP)	False Positive (FP)
	Negative (0)	False Negative (FN)	True Negative (TN)

Table 6.5 Common loss metrics for classification tasks.		
Name	Common equation	Ideal use case
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	Balanced count of all category labels. All categories are equally important but not the specifics about any single category.
Precision <i>aka</i> positive predictive value	$\frac{TP}{TP + FP}$	It is most important to get most positive predictions right, but false positives are costly.
Negative predictive value	$\frac{TN}{TN + FN}$	It is most important to get most negative predictions right, but false negatives are costly.
Recall <i>aka</i> sensitivity <i>aka</i> true positive rate	$\frac{TP}{TP + FN}$	It is most important that you are capturing all positive instances and it is costly to miss one.
Specificity <i>aka</i> true negative rate	$\frac{TN}{TN + FP}$	It is most important that you are capturing all negative instances and it is costly to miss one (e.g., it would be costly for someone to not be prepared for the occurrence of behavior).
F1 (generalizes to F β)	$(1 + \beta^2) \times \left(\frac{\text{Precision} \times \text{recall}}{\beta^2 * \text{Precision} + \text{recall}} \right)$	Detecting the occurrence of the phenomenon is most important and both precision and recall are equally important.
False positive rate <i>aka</i> Type I error	$\frac{FP}{FP + TN}$	The cost of raising a false alarm is very high and you want to avoid it.
False negative rate <i>aka</i> Type II error	$\frac{FN}{TP + FN}$	The cost of failing to detect the occurrence of the phenomenon is high.
Matthew's correlation coefficient	$\frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$	Imbalanced count across category labels. All classes equally important but not the specifics about any single category.
FN, <i>False negative</i> ; FP, <i>false positive</i> ; TN, <i>true negative</i> ; TP, <i>true positive</i> .		

we have a binary DV. Each metric has its strengths and weaknesses with researchers and practitioners typically using more than one metric to quantify how well a classification model performs. Lastly, as described in the section titled, “Classification models,” extending these out to situations where our DV has three or more potential labels is easy. You just create a confusion matrix modified for a one-vs.-rest analysis and then calculate the metric; or you create a confusion matrix for each one-vs.-one analysis, calculate your metric for each, and analyze the resulting set of loss metrics.

Accuracy

Accuracy is likely the classification loss metric that is most familiar to behavior analysts. Accuracy is calculated by simply counting up the number of times the model was “right” (true positive and true negative) and dividing by the total number of predictions. If multiplied by 100, accuracy is simply the percentage of predictions the model got right.

Accuracy is only a good metric when two conditions are met. The first condition is that you have a similar amount of both positive and negative category labels. To see why, consider a situation where 99 out of 100 times the behavior occurs. A very simple and noninsightful model would be to always predict the behavior will occur. In this situation, the model would have an accuracy of 99% but it would be hard to argue the model is telling us anything useful. The second condition wherein accuracy is useful is when we do not care about what kind of correct or incorrect predictions our model makes. As described above with [Table 6.4](#), our model can make correct and incorrect predictions in two ways each. Accuracy does not differentiate the two correct predictions from each other nor the two incorrect predictions from each other. To the extent that you have imbalanced category counts or that it matters you can differentiate correct/incorrect categories, accuracy is not the best metric to use.

Predictive value

Predictive values are a second set of metrics often used to quantify how well classification models make predictions (rows 2 and 3; [Table 6.5](#)). As the names imply, *positive predictive value* (aka precision) focuses on how well our model predicts the positive values (the “1”s) and *negative predictive value* focuses on how well our model predicts the negative values (the “0”s). In [Table 6.4](#), each predictive value metric focuses on one of the rows in the confusion matrix. Using the example above where behavior occurs on 99/100 trials, if it is really important that our model predicts that one instance where behavior does not occur, then negative predictive value captures that. Here, always predicting behavior will occur which would lead to 0% negative predictive value as it misses the one opportunity it had (i.e., $0/[1 + 0]$). This obviously isn’t the entire story as the model does well with positive predictions coming in at a 99% positive predictive value. As a result, most researchers using predictive value metrics would report on both positive and negative predictive values rather than either in isolation. And, if you have a model that has a really high positive/negative predictive value but not the other, then you know the model needs some improvement.

True rates

True rates are often used to quantify how well classification models capture the true rates that positive and negative labels occur (rows 4 and 5; [Table 6.5](#)). As the names imply, *true positive rate* (aka recall

aka sensitivity) focuses on the extent to which our model captures all positive instances that occurred, and **true negative rate** (*aka* specificity) focuses on the extent to which our model captures all negative instances that occurred. In Table 6.4, each of these focuses on one of the columns in the confusion matrix. These metrics are typically used when it would be a big deal to miss either a positive occurrence (e.g., a diagnostic test failing to detect cancer when it is present; a Registered Behavior Technician [RBT] failing to document that abuse was observed) or to miss a negative occurrence (e.g., failing to rule out disease with expensive and intensive treatment regimen; failing to rule out a behavioral function leading to wasted treatment resources). Similar to predictive values, true rates are often reported in conjunction with one another rather than in isolation.

F1 ($F\beta$)

Positive predictive value (precision) focuses on the top row of Table 6.4 and true positive rate (recall) focuses on the left column of 6.4. Given both of these focus on different aspects of positive occurrences, researchers often want to quantify how well classification models balance both predictive values and true positive rates. $F\beta$ scores do this by combining positive predictive value and true positive rate into a single, balanced metric. The most common way to do this is to give equal weight to each by setting $\beta = 1$. When $\beta = 1$ it is called the F1 score. However, β is a continuous number and, technically, you could swap it out for whatever positive real number you'd like (i.e., any value between 0 and infinity). β values greater than 1 give more weight to true positive rates (recall) and values between 0.0 and 1.0 give more weight to positive predictive value (precision). Common versions of $F\beta$ are F2 which gives twice as much weight to true positive rates and F0.5 which gives twice as much weight to positive predictive value. Lastly, if the absence of the phenomenon was more important to your research or practice context, you could calculate $F\beta$ scores using negative predictive value and true negative rates.

False rates

As the name implies, these metrics tell you how often your model makes an error out of all the opportunities it had to make an error. If you have taken any kind of statistics class in your life, you likely have heard of Type I and Type II errors. Those are just other names for false rates. Type I errors (***aka false positive rate***) quantify how often

our model incorrectly predicts behavior would occur relative to all opportunities wherein it could have made this kind of error. Type II errors (*aka false negative rate*) quantify how often our model incorrectly predicts behavior would not occur relative to all opportunities wherein it could have made this kind of error.

False positive and false negative rates are typically used when the cost of making an error is much greater than the benefit of getting predictions right. This is why Type I and Type II errors are often used when analyzing experimental results. To err on the side of caution, scientists want to make sure the likelihood is low that they make an error (i.e., claim an effect exists when it actually may not). For classification modeling, specifically, researchers will often use false positive and false negative rates in conjunction with either positive/negative predictive values or true positive/negative rates. This allows scientists to get a balanced look at how well a model is capturing all four possible model outcomes.

Matthew's correlation coefficient

Matthew's correlation coefficient (MCC) is a great single metric that is an alternative to accuracy for looking holistically at the performance of classification models. Though the mathematical reasoning behind MCC is well beyond the scope of this book, MCC has shown to be quite the robust metric. Specifically, MCC is a more reliable loss metric than accuracy and F1 score; MCC is high only when predictions are good for all four categories in the confusion matrix; and MCC performs well with imbalanced datasets ([Chicco & Jurman, 2020](#)). We saved it for last so that you would have a good sense of the limitations of accuracy, you would understand the benefit of focusing on individual categories in the confusion matrix, and you would know what an F1 score is and its limitations. There still is significant benefit to reporting on multiple metrics as is common with many of the metrics above. However, if you want to use only one metric to quantify how well your model performs and that balances all four categories, MCC is likely the best first choice.

Loss metric summary

To reorient ourselves in the landscape of this chapter, we start by identifying what kind of model is best suited to our IV-DV context. From there, we let computers do their thing and fit the model to our data so that we can describe it or make predictions about future collected

data. Next, we examine how good our model actually is by converting our residuals or confusion matrix counts into a loss metric wherein the chosen metric is determined practically by the DV and how we want to communicate about it afterward. The value of the loss metric allows us to determine whether we can move to the final stage of model evaluation.

What counts as a “good” model depends on the loss metric and model type. For linear regression tasks, r^2 values greater than 0.85 (experimental contexts) or 0.60 (applied/complex contexts) are typically considered “good.” For nonlinear regression models, MAE and RMSE are often used because it keeps the metric in the units of our DV. Here, “good” is determined pragmatically. For regression model comparisons, AIC and BIC are better choices because they evaluate model complexity and model fit simultaneously. AIC and BIC, however, are relative measures (i.e., they can’t really be interpreted in isolation) with lower values meaning better models.

Lastly, a suite of loss metrics exist for classification models. Scientists will typically pick a pair of metrics based on what is most important to their context. These include the model’s predictive capabilities (predictive value), how well it captures true rates of each category, or the types of errors the model makes. Sometimes researchers want a single metric that blends either a pair of metrics (F/β) or captures all four correct/incorrect possibilities in a single number (accuracy, MCC). Though accuracy is likely the most familiar to behavior analysts, it is an appropriate metric in only specific circumstances and past research suggests MCC is the best single metric for balancing all four correct/incorrect possibilities.

How influential is each independent variable on my dependent variable?

If you have a good model fit, congratulations! You have successfully taken a suite of measured IVs, some likely controlled and some likely not; and you have found a way to quantify the precise relationship between those IVs and your DV within the environmental contexts they occurred. This is no easy feat! You now get to enjoy the deliciousness of which this entire enterprise is about in learning about the relative influence of each of those IVs on your DV. You can do this by way of evaluating the values that were assigned to each free parameter

in your model. And, likely unsurprising to you at this point in the chapter, what exactly those free parameter values mean depends on the type of model we used.

Refresher on free parameters

A free parameter is any variable in our model that we did not measure directly and we did not have a specific set of values for in our dataset. For example, consider the classic equation for delay discounting:

$$V = \frac{A}{1 + kD}. \quad (6.4)$$

In this nonlinear regression model, V is the value¹³ of a larger later reinforcer at the specific delay of D ; A is the measured amount of the reinforcer (e.g., \$100, 10 minutes of iPad, 10 pieces of candy, 10 seconds of tickles); and k is the rate whereby value decreases as a function of increasing delay. In delay discounting experiments, we control for the amount of the reinforcer (A) and the delay (D) to when the participant would contact A . We also directly measured the choice that participants made (our DV) that equated to the value (V) of the reinforcer amount A at time point D (i.e., larger later reinforcer). Thus V , A , and D are all “fixed” values in our dataset—we cannot change those measurements when we fit the model as the data has already been collected. When we fit Eq. 6.4 to our data, the only variable that is “free to vary” per participant is k . Thus k is a free parameter. And, a participant’s k value tells us the influence of increasing delay (our IV) on reinforcer value (our DV) for that participant.

One final note on terms. Different areas of science sometimes use different terms to refer to the estimated values for each free parameter. Names you are likely to see in the wild include model coefficients, variable coefficients, coefficients, beta weights, variable weights, feature weights, feature importance, and variable importance. At a high level, they all mean essentially the same thing: the estimated value of a free parameter in our model that is meant to capture the relative influence of that IV on the DV.

¹³ By value, we just mean the amount of behavior that the reinforcer is likely to maintain. Typically “behavior” is made even more specific to the context such as the duration, intensity, rate, latency, or relative allocation.

Interpreting linear regression parameters

Interpreting linear regression model coefficients is the easiest of all the model types. Quite simply, the estimated free parameter in a linear regression model describes how much your DV will change for each one unit increase in the IV. If the coefficient is positive, then it means that every one unit increase in your IV leads to an increase in your DV by the amount of the coefficient. And, if the coefficient is negative, then it means that every one unit increase in your IV leads to a decrease in your DV by the amount of the coefficient. When you have more than one IV in your model, the coefficient represents the isolated influence of that IV on your DV (i.e., holding all other IVs constant). Importantly, most statistical outputs will provide you with a confidence interval for the estimated value of the free parameter and a p value. If the confidence interval contains 0.00 or the p value is greater than 0.05, it is unlikely that your IV has a consistent influence on the DV for that participant.

Interpreting nonlinear regression parameters

Well, that was the end of the easy stuff. Now things get a bit more...messy. Let's consider again two common nonlinear models in behavior analysis in discounting and demand (Fig. 6.3). When we fit these models to the data, we can interpret the parameters kind of similar to above. For every one unit increase in our IV, the DV changes by the amount and in the direction of the estimated free parameter. However, the amount and direction changes based on the structure of the nonlinear model so it's not quite the easy interpretation we get with linear models.

Let's use the delay discounting equation as an example of the complexity (Eq. 6.4). The structure of this model is called a "hyperbolic" model because of the type of curve it produces (Fig. 6.3). At small delays, a one unit change in delay (e.g., increasing the delay from one to two weeks) leads to a large reduction in reinforcer value compared to a one unit change at larger delays (e.g., increasing the delay from 10 years to 10 years and one week). You can see how interpreting an estimated k value of 0.8 gets messy. How much does reinforcer value change with every one unit increase in delay? Well, it changes by 0.8 hyperbolic units of reinforcer value (in the mathematical sense not the exaggerated sense). For most, this is not intuitive to "see" without an accompanying graph. But, that's okay because k is still a precise

statistical estimate that allows for quantitative descriptions and predictions of behavior. Thus we can practically use k values for within-subject comparisons of discounting with different commodities (e.g., [Baker et al., 2003](#)), between-subject comparisons of discounting in different clinical populations (e.g., [Baker et al., 2003](#)), and to predict the likelihood that an intervention will be successful for any one participant (e.g., [Dallery & Raiff, 2007](#); [Reed & Martens, 2013](#)).

Returning to a broad generalization, the estimated parameters in nonlinear regression models tell us the amount and direction our DV changes relative to the model's structure. If we have multiple IVs in our model, then the estimate tells us the relative influence of that IV holding all others constant. Lastly, like all statistical estimates covered in this book, we can practically use that quantitatively precise estimate of the influence of our IV on our DV to describe and predict behavior toward an answer to a research or practice question.

Interpreting classification model parameters

Interpreting the estimated parameters from classification models is also a bit messy, technically but has a high-level interpretation that is easy to grasp. Let's start with the easy bit. As described earlier in this chapter, the output of most classification models is a probability that behavior will fall into a specific category. If we have two categories, it is the probability behavior will fall into whatever we defined as a "1." If we have more than one category (e.g., one-vs.-rest), then it is the probability that behavior will fall into whatever "one" category we are focusing on. Thus, at a high level, you can think of the estimated parameters in classification models as telling you how much a one unit change in our IV will change the probability that behavior will fall into category "1." Thus greater parameter estimates suggest that IV increases the probability more than lesser parameter estimates. Also similar to above, if we have multiple IVs, the parameter estimates provide you with the relative influence of that IV on your DV holding all others constant. Easy enough, right?

Now the messy part. Kind of like nonlinear regression models, the parameter estimates that your computer spits at you are in units that depend on the type and structure of the model. What this means is that you have to use a specific equation to convert the parameter estimates into the actual change in probability that your IV has on your DV. Fortunately, Google is your friend here. Simply search, "How to

convert [model] coefficients into probabilities?” and you will likely find the website you need to help you on your quest. Though this requires a bit of extra work, those that partake will be rewarded with parameter estimates that are much easier to interpret and you will better understand the precise influence of each IV on the probability of observing your DV.

Interpreting relative influence in multivariable models

We will close this section on interpreting parameter estimates by discussing how to interpret the relative influence of multiple IVs on your DV. In most multivariable modeling situations, we are interested not just in the size of the influence that each IV has on our DV, but we want to know which IVs are more important or more influential than the other IVs. To do this requires that we transform all of our IVs into a similar scale *before* we fit the model to our data.

To demonstrate why we need to do this, consider a situation where we want to know the precise quantitative influence of reinforcement schedule and reinforcer amount on rate of responding. In this situation, the parameter estimates from our regression model will be in the units of reinforcement schedule and reinforcer amount. Comparing parameter estimates of, say, 2.0 for fixed response (FR) requirement changes and 1.0 for the number of tokens is a bit like comparing apples and oranges (well, technically, schedules and tokens). It's unclear how to easily do that to answer a question about whether schedule requirements or reinforcer amount have a greater influence on response rates. But, never fear, smart people throughout history have solved this challenge for us. The most common method is to use data transforms to get our IVs on the same page.

A quick Google search for “methods of feature scaling” will likely return to you a very large list of ways you can go about doing this. For example, min-max scaling converts your IVs into the range of 0.00–1.00 corresponding to the min and max values of your IV, respectively. And, standard scaling (*aka* converting to a *z*-score) converts your IVs into their location on a normal curve (see Chapter 3¹⁴). Such *feature scaling* is useful because now all our IVs are on the same scale in terms of the size they can take. The benefit is we can now compare the parameter estimates to one another and obtain a precise

¹⁴ A note of caution: This transformation only makes sense if your IV is normally distributed.

estimate of how much more of an influence one IV has on our DV compared to another. For example, had we “feature scaled” our IVs in the previous paragraph first, parameter estimates of 2.0 and 1.0 for FR schedule and number of tokens, respectively, would let us know that FR schedule has twice the influence on rates of responding as the number of tokens we use with each reinforcer delivery. As a result, you could use this information to more precisely predict how changing each of these IVs will change behavior.

There is a downside to feature scaling. Because we have converted the units of our IVs out of their original form, our estimated parameters no longer are in the units of our IVs but in the transformed units. As a result, if you want to interpret the parameter estimates as we described earlier while also interpreting the relative influence of each parameter, you can always fit two models: once with the IVs in their raw, original values and once with the scaled IVs.

Chapter summary

In many research and practice situations, we are likely to find ourselves with a set of data on one or more IVs and one or more measures of behavior (DVs). Experimentation is obviously the ideal method to determine exactly how one or more IVs influence behavior, it is not always possible nor practical. In these situations, we may still want to get a sense for how different experimentally and nonexperimentally controlled variables influence behavior. Modeling helps us do that.

The method and approach to modeling we take depends on several factors. These include (1) whether our DV is continuous or discrete; (2) whether we have one or more IVs; and (3) the assumed linear or non-linear relationship between our IVs and DV. Once we build a model, we evaluate how good it is by analyzing loss metrics. Loss metrics also are chosen based on several factors. These include (1) whether our DV is continuous (regression models) or discrete (classification models) and (2) what we want to emphasize in the analysis (i.e., the units of our DV or model comparisons for regression; specific cell(s) in the confusion matrix for classification).

Lastly, assuming we have a model that describes our data well or makes accurate predictions, we can interpret the estimated parameters of our model. Typically, these tell us the relative influence each IV in

our model has on our DV. Modeling with the raw IV values allows us to interpret the influence of one unit changes in our IV on our DV where the DV changes based on the structure of our model (e.g., linearly, hyperbolically, exponentially). However, this approach does not provide us with comparable estimated parameters as our IVs are often in different units. Modeling only after feature scaling solves this problem and allows us to make claims about how much more of an influence one IV has on behavior compared to other IVs.

To close, if you understood everything in this chapter summary then a sincere and tremendous, “Bravo!” We appreciate this material is a bit dense and, as mentioned before, modeling deserves a book-long treatment in its own right. Nevertheless, we hope this introduction at least whets your whistle and you can begin your journey toward controlling for nonexperimentally controlled variables and making sense of the beautifully complex world we live in.

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